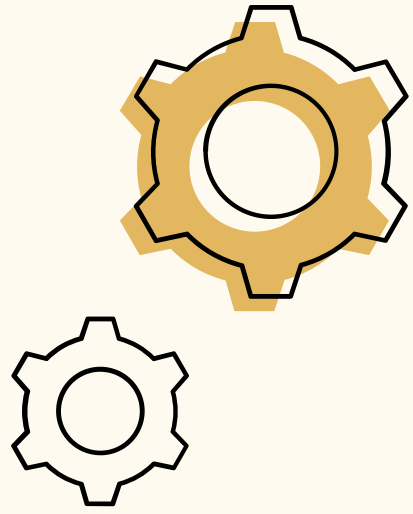


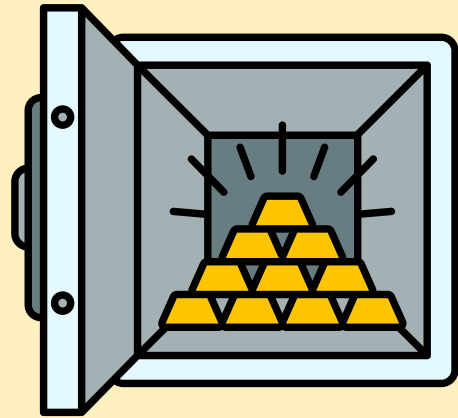
Machine Learning-Based Price Trend Prediction and Automated Trading System for Gold

Lim Jia Xuan





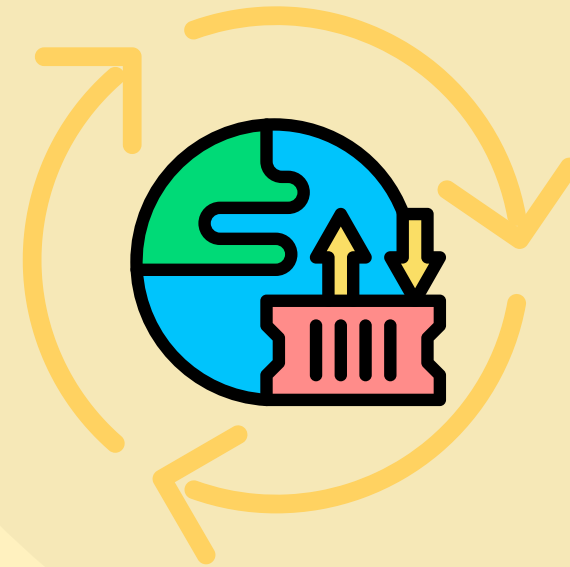
INTRODUCTION



Safe-haven asset



**Strong long-term
price growth**



**Counter-cyclical
during crises**



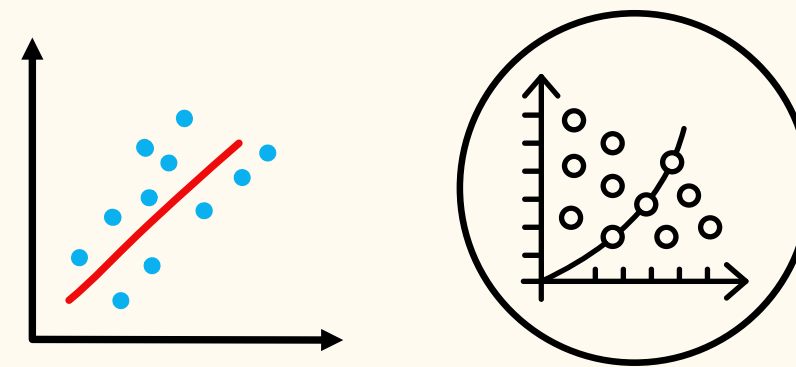
**Influenced by macro
& geopolitics**

MOTIVATION



Traditional models fail in volatility

- Linear assumptions fail in non-linear, unstable gold markets



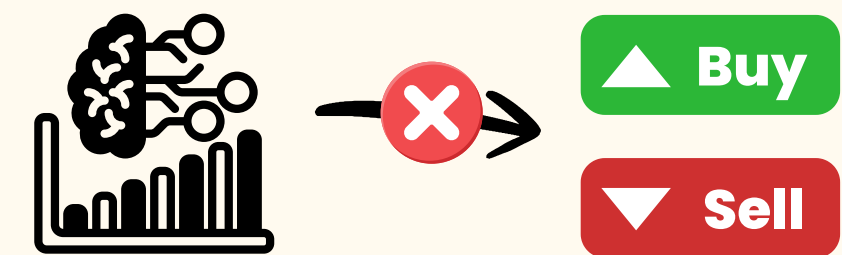
Manual decisions cause latency

- Trades manually may introduces decision delay



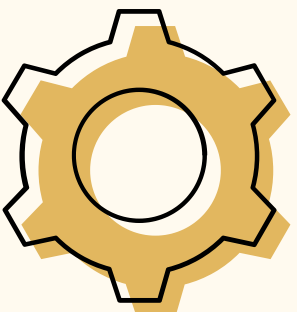
ML predictions not linked to trading

- No direct mechanism to convert predictions into trading actions

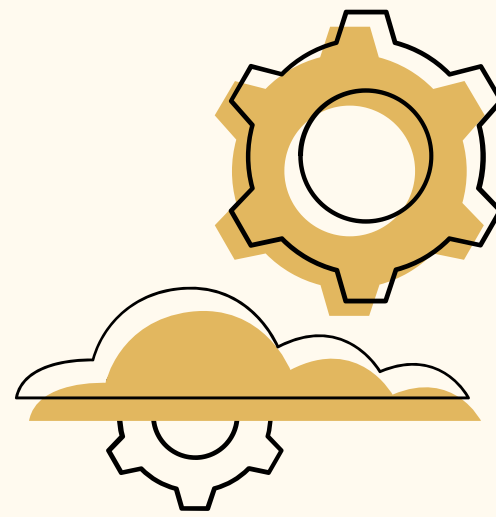


News sentiment often ignored

- Most models use only historical data, ignoring news-driven market signals



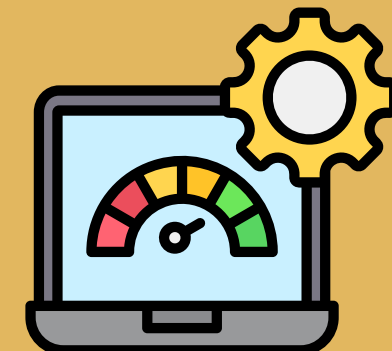
OBJECTIVES



Develop a machine learning framework for predicting gold price movement

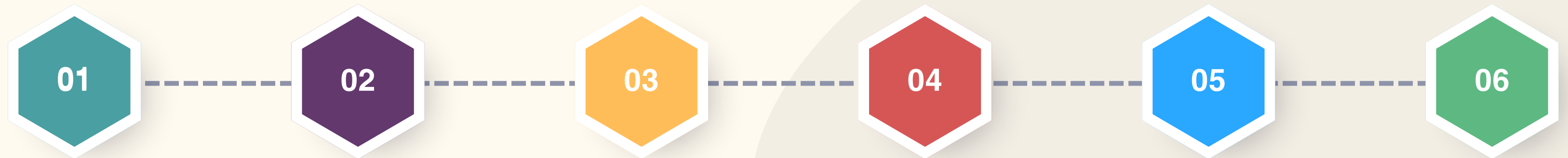


Integrate financial news sentiment into the predictive and decisionmaking process



Design and evaluate a risk-aware automated trading simulation based on model-driven signals

METHODOLOGY



Business Understanding

- Predict gold price direction
- Integrate news sentiment
- Integrate Dashboard



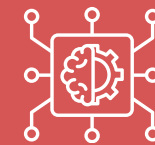
Data Understanding

- Hourly gold & macro prices
- Financial news headlines
- Sentiment & market regimes



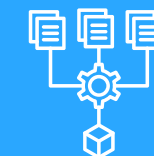
Data Preparation

- Hourly time alignment
- Feature engineering & lagging
- Noise & bias removal



Modeling

- Multiple ML base models
- Soft-voting ensemble
- Calibrated probabilities



Evaluation

- Precision, Recall, F1
- Return, Sharpe, Drawdown
- Threshold behavior



Deployment

- Modular microservices
- Simulation-only execution
- Transparent dashboard

Tools & Technologies

Core Development



Python

Machine Learning



scikit-learn (LR,
SVC, RF, GB,
MLP)

XGBoost

XGBoost

NLP



FinBert

Data Sources



Yahoo Finance



Google News
RSS

Data & Feature Processing



Pandas

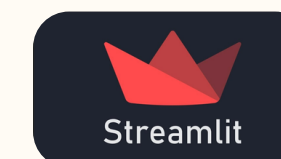


NumPy



TA-Lib

Visualization & Dashboard



Streamlit

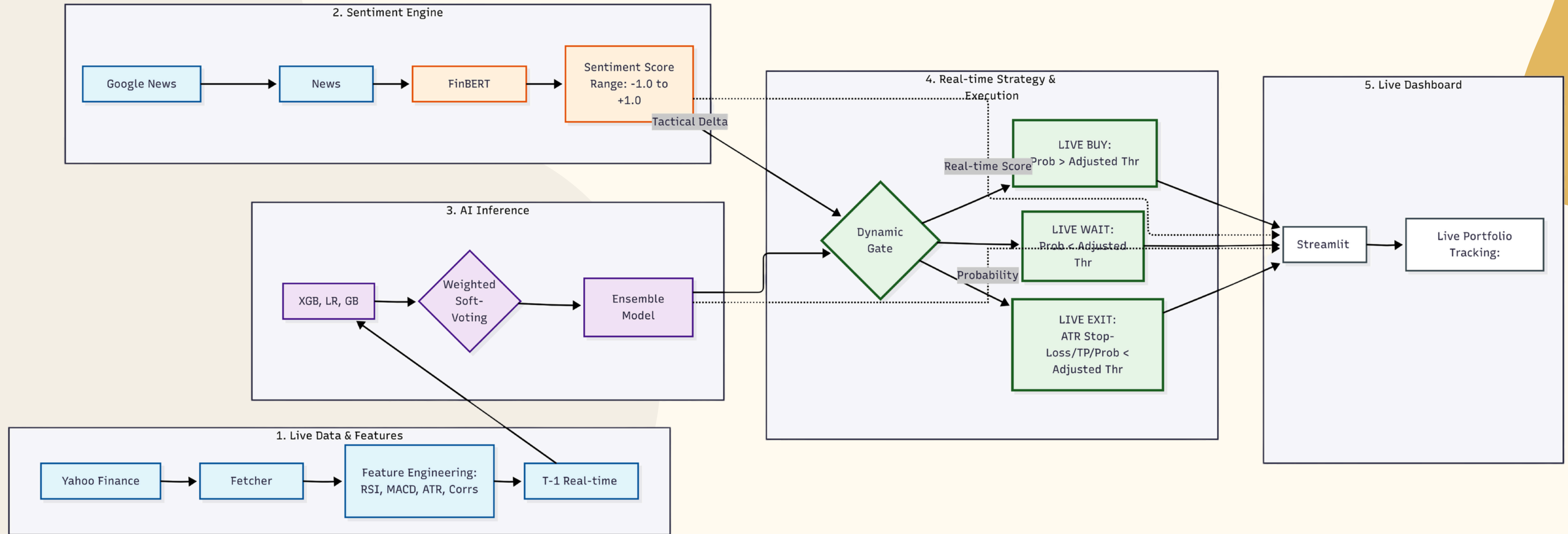


Matplotlib

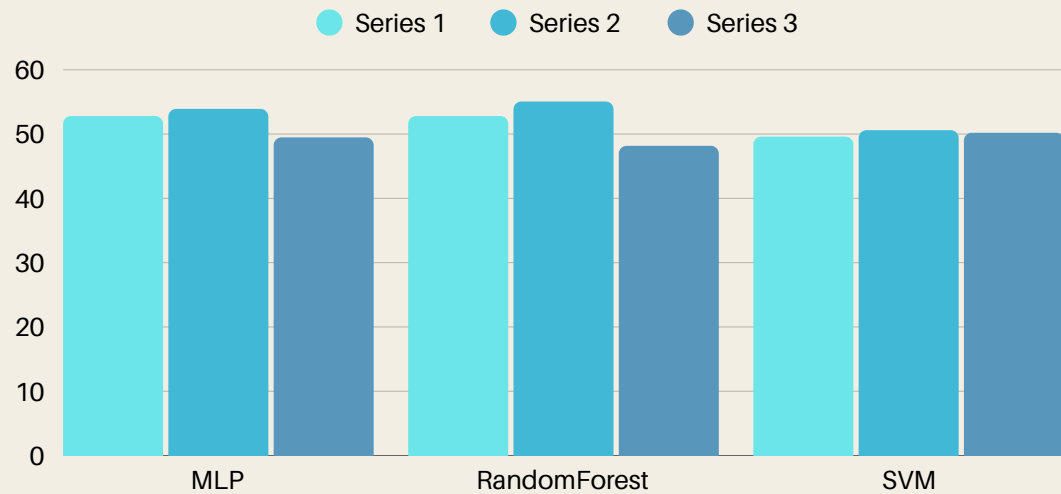
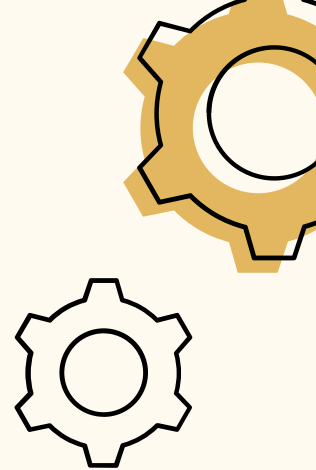


Plotly

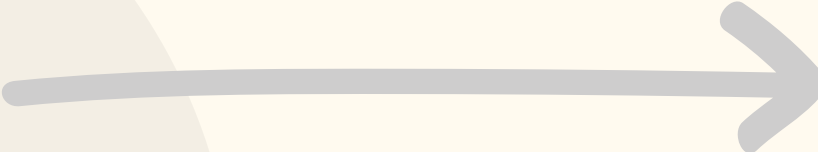
System Architecture



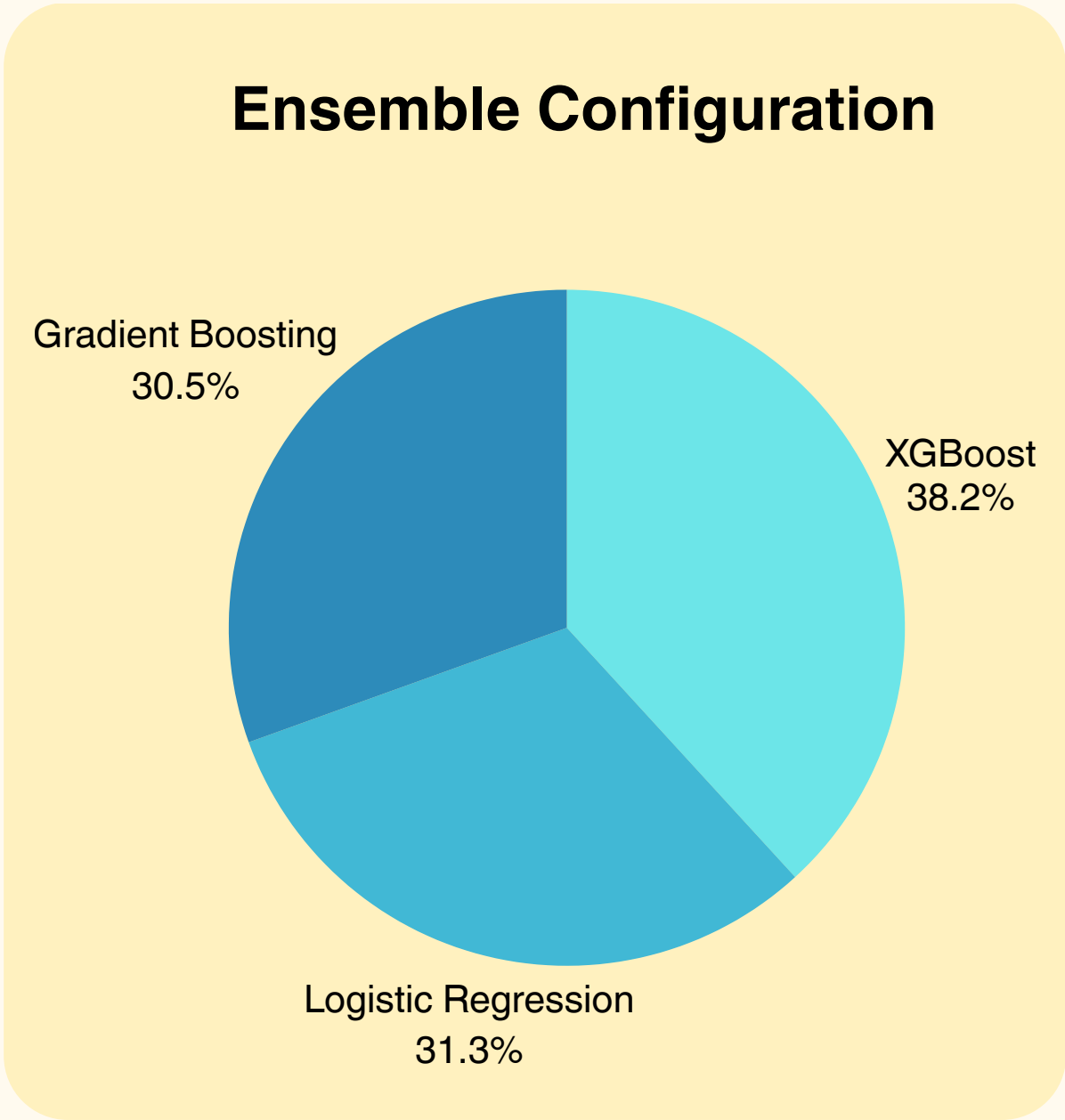
Ensemble Model



Smart Score = 0.7 × Precision + 0.3 × CV Accuracy

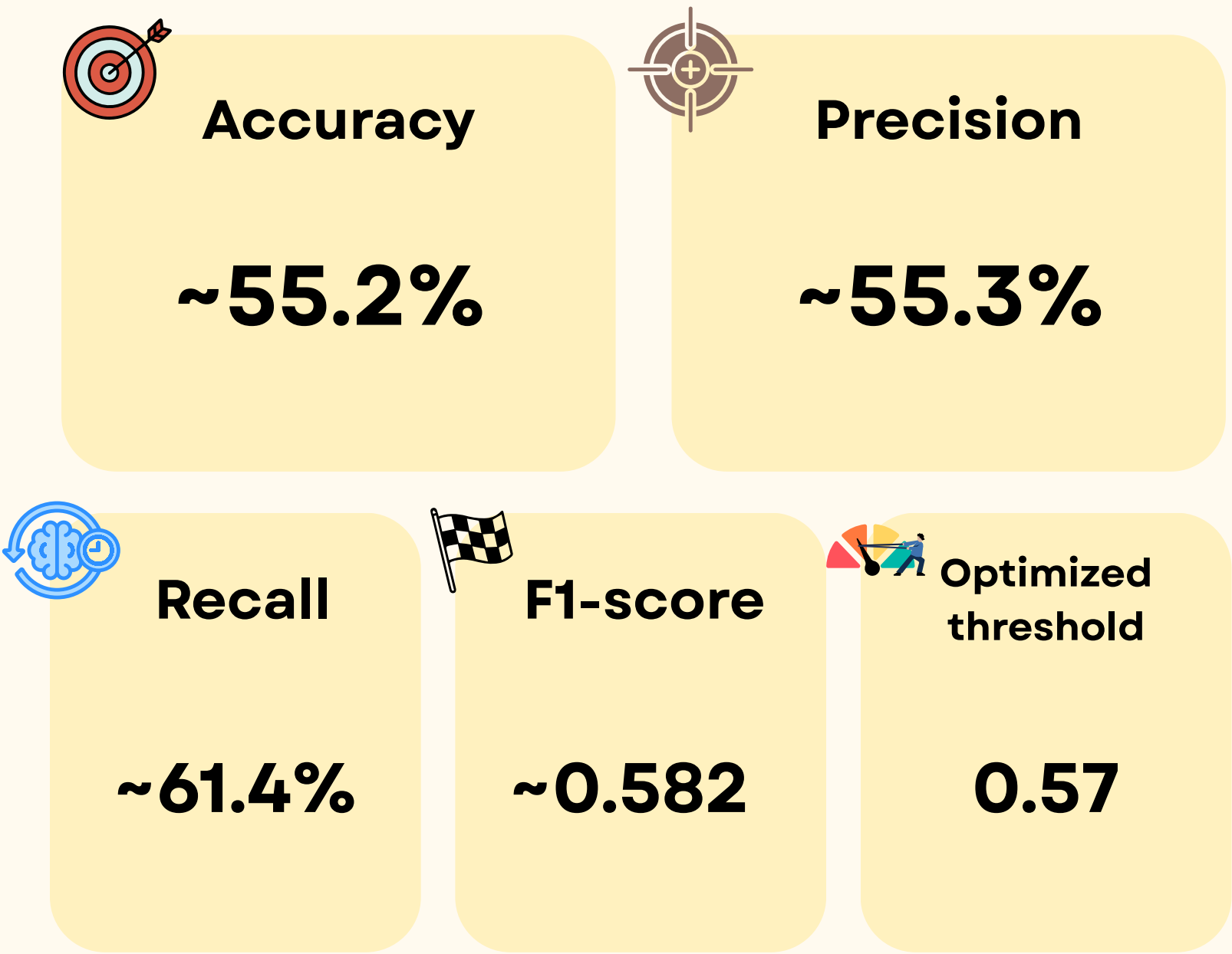


TOP 3 MODELS



Ensemble Model Performance

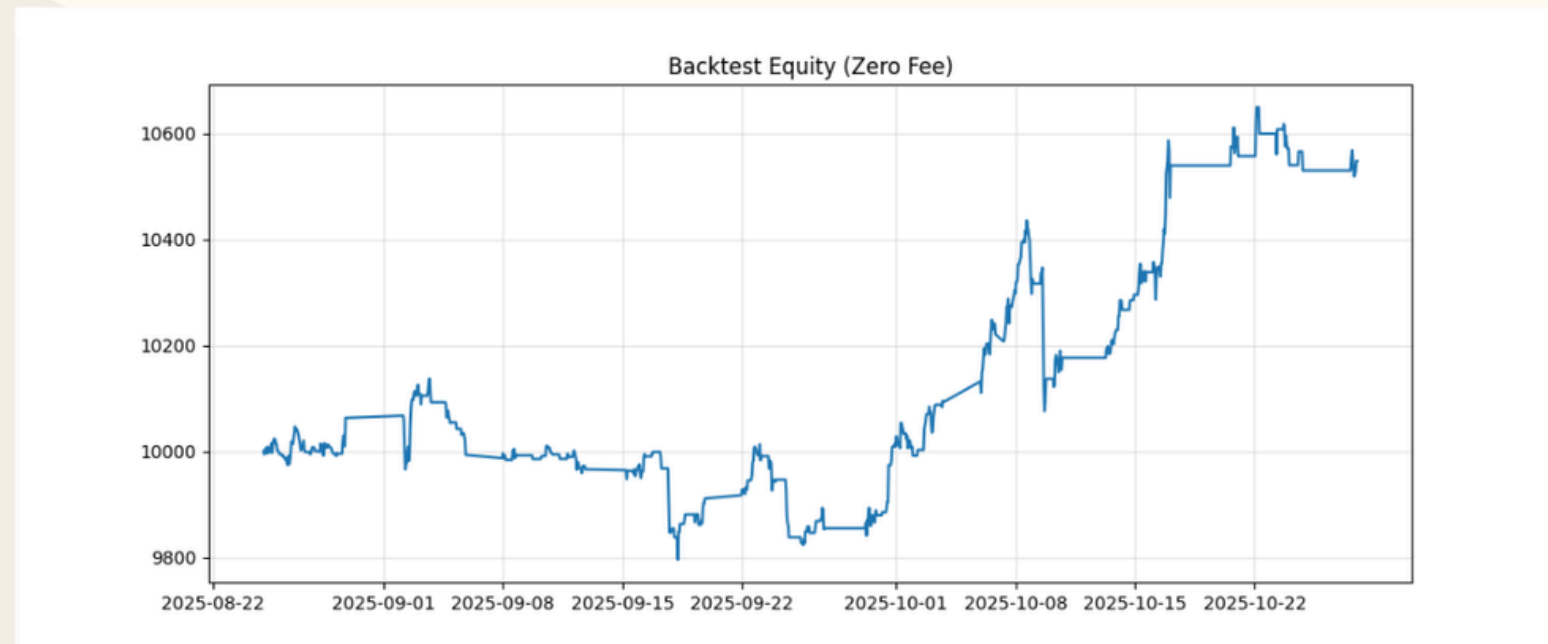
Predictive Performance



Top Predictive Drivers

Trend & Momentum (MA Difference, RSI (14), Lagged Return (1h), Momentum Filter)	0.328
Macro & Cross-Asset (Gold / USD Ratio, Gold / BTC Ratio, USD Correlation (24h))	0.319
Volatility & Risk (VIX Index, ATR (14))	0.200
Temporal Structure (Hour_Sin, Hour_Cos)	0.153

Trading Performance (Out-of-Sample)



Equity Curve

1	side	entry_time	entry_price	units	sl_price	tp_price	entry_com	entry_prob	exit_time	exit_price	exit_comm	pnl	exit_reason
12	long	2025-08-28 18:00:(	3475.899902	2.589258	3452.221	3511.418	0	0.563883	2025-08-2	3471.4	0	-11.6517	Strategy Exit
13	long	2025-08-29 06:00:(	3470.600098	2.593211	3456.714	3491.429	0	0.54917	2025-08-2	3470.7	0	0.258941	Strategy Exit
14	long	2025-08-29 15:00:(	3504.100098	2.56842	3486.572	3530.393	0	0.524569	2025-09-0	3530.393	0	67.53093	TP
15	long	2025-09-02 04:00:(	3564.300049	2.52504	3540.043	3600.686	0	0.571299	2025-09-0	3540.043	0	-61.2501	SL
16	long	2025-09-02 07:00:(	3555.800049	2.531076	3520.571	3608.643	0	0.549429	2025-09-0	3596.6	0	103.268	Strategy Exit
17	long	2025-09-03 00:00:(	3595.699951	2.50299	3560.493	3648.511	0	0.550079	2025-09-0	3596.9	0	3.003465	Strategy Exit
18	long	2025-09-03 05:00:(	3602.800049	2.498057	3569.479	3652.782	0	0.524569	2025-09-0	3601.6	0	-2.99755	Strategy Exit
19	long	2025-09-03 15:00:(	3624.5	2.483101	3592.207	3672.939	0	0.527294	2025-09-0	3619.5	0	-12.4155	Strategy Exit

Trade Logging

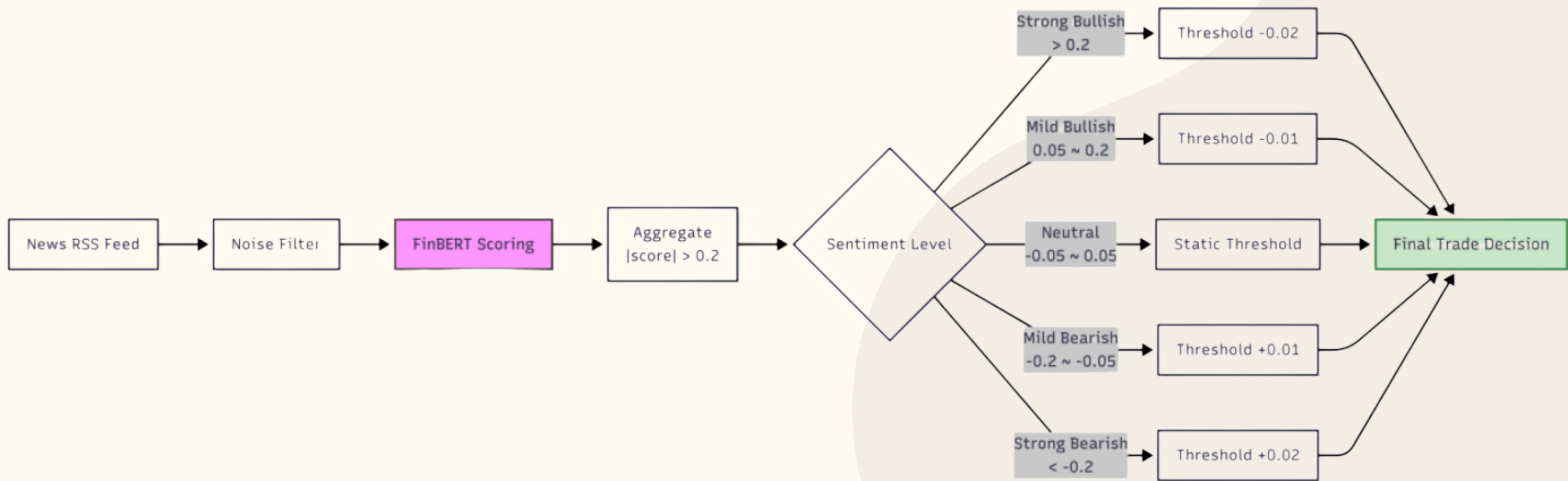
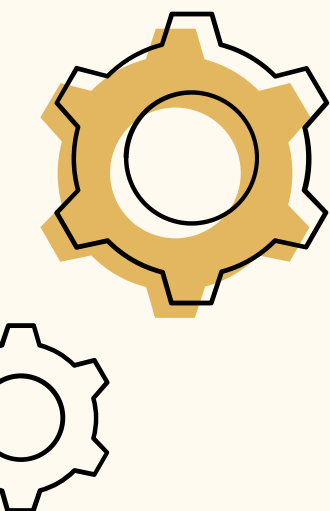
Trading Strategy

STOP LOSS = ATR * 3.0
TAKE PROFIT = ATR * 4.5

ATR = Average True Range

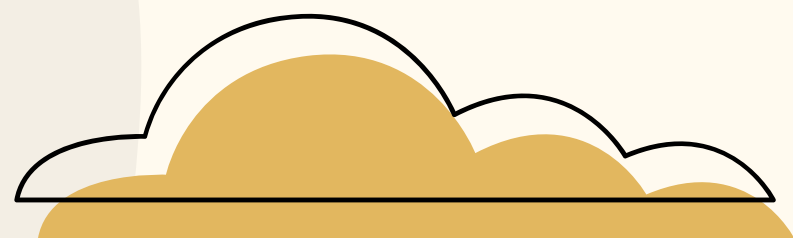
Total Return	+5.48%
Sharpe Ratio	3.05
Win Rate	58.0%
Max Drawdown	-3.45%
Total Trades	119

Sentiment-Aware Decision Making

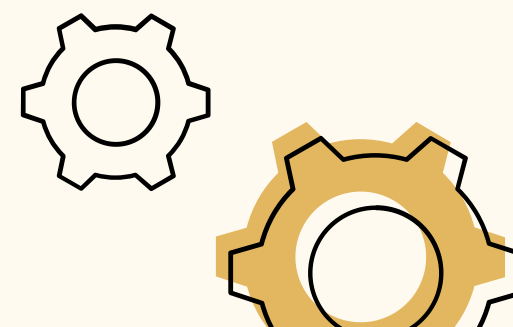


Date	Source	Title	Label	Score	URL						
2025-12-19T04:00:00Z	Forbes	The Geopolitical Risks Tha	Negative	-0.9994	https://news.google.com/rss/articles/CBMirgFBVV95cUxNQ2cyd3Z						
2025-12-19T03:37:00Z	Barron's	Stock Markets May Have B	Negative	-0.9504	https://news.google.com/rss/articles/CBMimwNBVV95cUxPZzd3W						
2025-12-19T03:28:08Z	Reuters	Gold eases on dollar stren	Positive	0.9692	https://news.google.com/rss/articles/CBMiqAFBVV95cUxQdjhRYU						
2025-12-19T03:02:43Z	TipRanks	Gold Holds Course Towarc	Neutral	0	https://news.google.com/rss/articles/CBMimAFBVV95cUxPOG5jSX						
2025-12-19T03:00:00Z	Geopolitical F	The Enduring Wedge Betwe	Neutral	0	https://news.google.com/rss/articles/CBMigwFBVV95cUxQMzlwTX						
2025-12-19T02:55:16Z	FXStreet	Can Gold carry the bullish	Positive	1	https://news.google.com/rss/articles/CBMi1gFBVV95cUxPT3lzSnFa						
2025-12-19T02:46:15Z	Bloomberg.cc	Gold, Silver Near Record H	Positive	0.9835	https://news.google.com/rss/articles/CBMiswFBVV95cUxPeTBhMU						
2025-12-19T02:45:00Z	The Motley Fo	Wall Street Has a Federal F	Negative	-1	https://news.google.com/rss/articles/CBMimAFBVV95cUxNN18yV						
2025-12-19T02:27:58Z	MSN	Gold price today: Rates dri	Negative	-0.9705	https://news.google.com/rss/articles/CBMi8AFBVV95cUxNQW5zU						

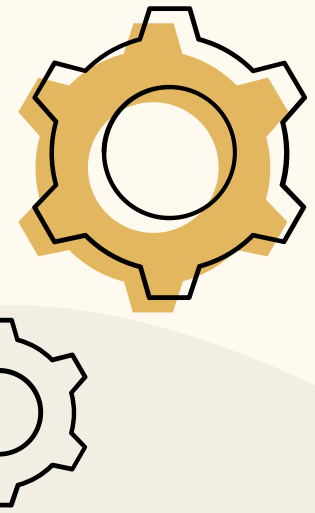
Latest News Headline



Demo Sessions



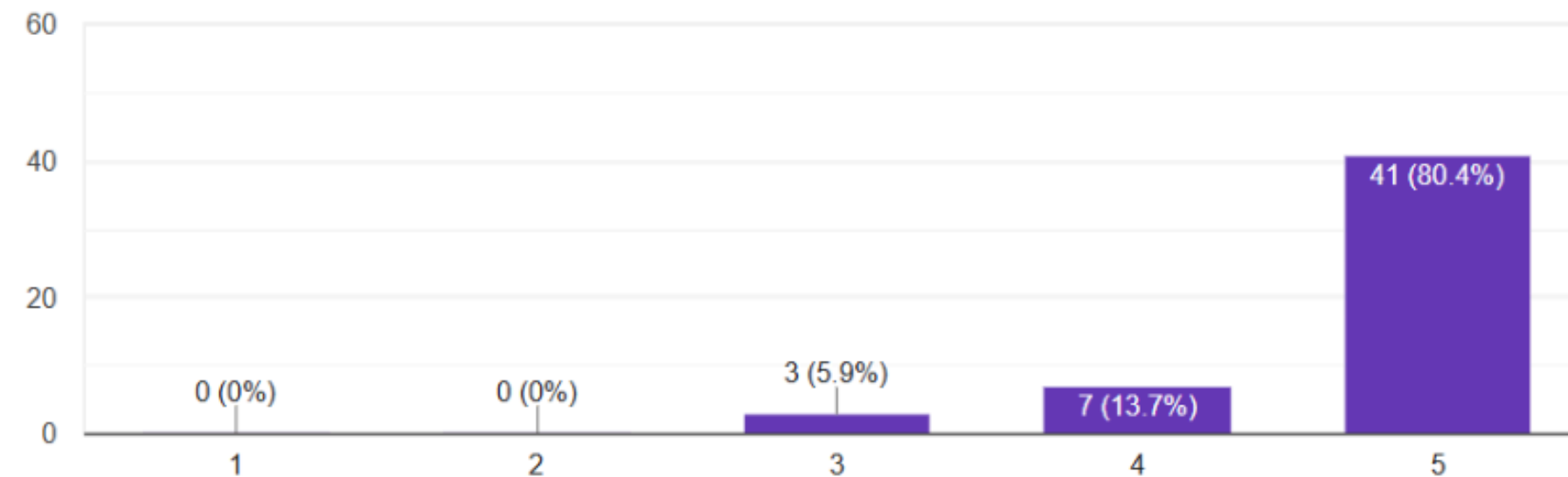
EVALUATION



Overall, I am satisfied with the performance and features of this AI trading system.

 [Copy chart](#)

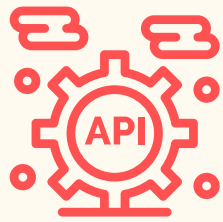
51 responses



Achieved Requirements

- All-in-one AI trading platform
- Automated data & news integration
- Clear UI with explainable AI
- High user trust & satisfaction

FUTURE IMPROVEMENT



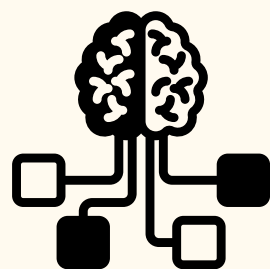
Integrate live economic
calendar APIs



Gold-Specific Sentiment
Intelligence



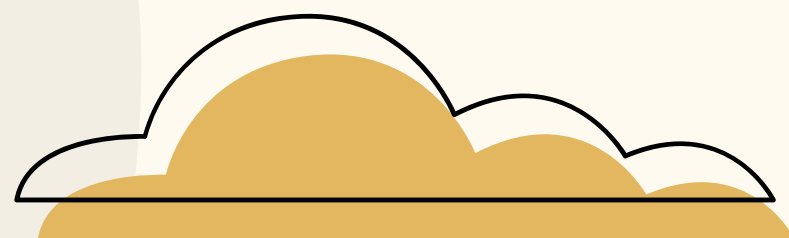
Cloud-based services



Sequential Market Modeling
(LSTM, N-BEATS)



Live Trading & Execution
Realism



THANK YOU